Dancing Stellar Streams in Merging Dark Matter Halos

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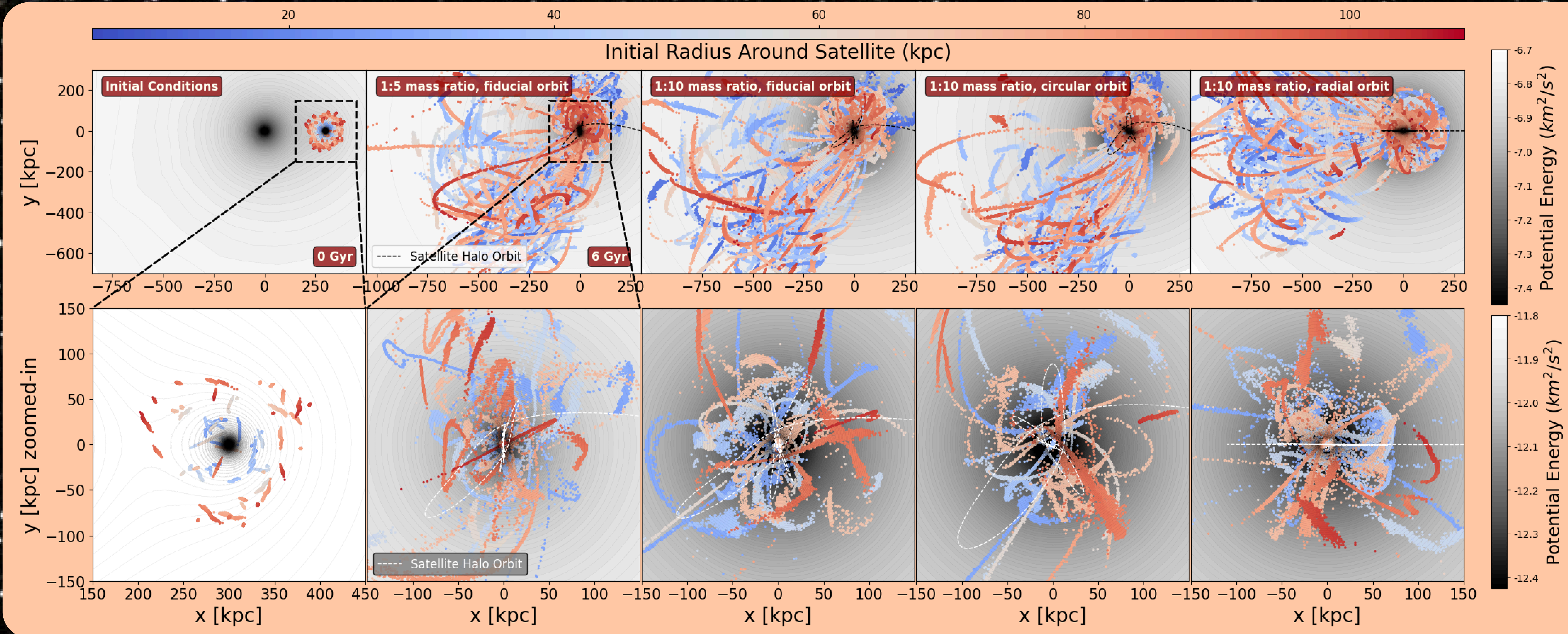
Background and Methods

Thin stellar streams form from the tidal stripping of star clusters, where escaped stars have closely similar orbits around a host galaxy. This dynamically cold debris provides clues to its star clusters' past evolution, insight into the host galaxy's formation history, and a measurement of the mass distribution in the host's dark matter halo (e.g. Helmi 2020, Johnston 2016). *The Large Magellanic Cloud (LMC) is currently merging with the Milky Way (MW) and could be bringing in its own set of globular clusters and streams.*

In this poster, we analyze simulations of stellar streams in an LMC-like satellite halo merging with a MW-like halo (parameters from Garavito-Camargo et al. 2019). The streams were formed around the satellite galaxy in a static halo potential using the Gala package (Fardal 2015, Price-Whelan 2017), a particle-spray approach. The satellite and host galaxy halo were then modeled and the streams were evolved during the galaxy merger using the simulation N-body code Gadget-4 (Springel 2021). Results presented here include a fiducial satellite orbit (0.27 eccentricity), a circular orbit, or a radial orbit, with either a 1:5 or 1:10 mass ratio between the satellite halo and the host galaxy halo.

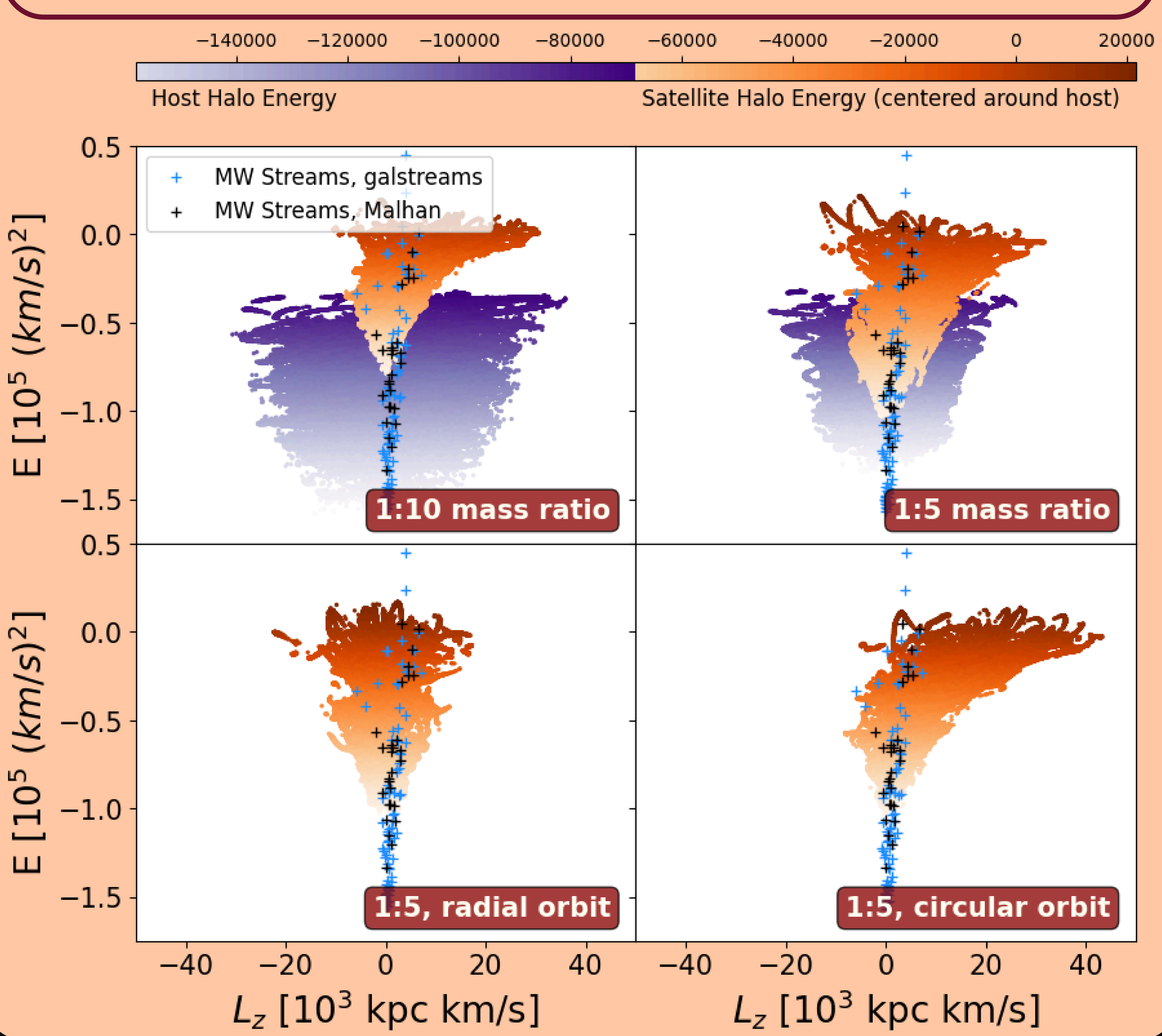


1. Stellar streams evolved in an accreted LMC-like halo

The initial and final distribution of low-mass streams, colored by their initial radius around the satellite. First column: initial conditions around the satellite; Second through fourth column: post-merger distributions for different mass ratios and orbits. The potential is in grey contours and the dotted line depicts the orbital path of the satellite halo as it merges. The top row shows all streams, while the bottom row zooms in on a subset of streams at smaller radii. Stellar streams from accreted satellites tend to keep coherent positions and velocities. Some streams can end at very large radii or even escape the host, while others may appear similar to streams formed around the host itself.

2. Energy vs Angular Momentum

The simulated host halo (purple) and satellite halo (orange) streams with the midpoints of observed MW streams from Malhan et al. 2022 (black) and galstreams (Mateu 2023; blue) plotted on top. Angular momentum is found from the observed proper motions and energy is calculated assuming a Hernquist potential and bulge, an MN3 approximation for an exponential disk, and an NFW Halo put into galpy's 2022 Milky Way Potential. Our simulations only account for a halo. Observed streams are narrower in Lz, which could be explained given two reasons: the real MW and LMC are not as far along in the merging process as our simulations have been evolved. Additionally, stream-like structure at such high angular momenta may be more challenging to observe in the Milky Way.



Main Findings

- Stellar streams from accreted dwarf galaxies largely stay coherent and while some are launched to large radii during the merger, others may appear like regular host halo streams.
- Observed streams in the Milky Way primarily lie within the host halo energy range, but a subset of streams populate the energies dominated by the accreted satellite streams
- When studying an individual stream, it is interesting that the stream seems to split into two in the xy-plane, with both parts still orbiting on approximately the same orbital plane. The split appears to depend on their accretion history as parts of the same stream can be stripped from the satellite during different pericenter passages.

Future Work

- Look for patterns between the initial properties of a stream/halo and final conditions around the host halo.
- Analyze stream morphology, orbit, thickness, phases of the orbit, etc.
- Look at the large changes in potential and kinetic energy
- Run more simulations with varying mass ratios, orbits, etc.
- Compare to stellar streams evolved in the host halo, in both isolation and a merger event

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3. Single Stream Analysis

Final conditions of a stream's cartesian coordinates, orbital coordinates, and energy and angular momentum, colored by their initial energy around the satellite halo. Stars with more negative energy (orange) start closer to the satellite's center, while stars that are less negative (purple) start further away. Post merger, the stars are no longer in a recognizable single stream-like structure and are now in their own individual orbits around the host halo. The stream seems to split in two, with the more defined section at a larger radius and therefore corresponding to the more-defined higher angular momentum stars.
The bottom two panels show the orbit of the stars around the host halo along with their energies, with the satellite halo's orbit plotted in black. Peaks and dips in the energy show where the satellite reaches its apo- and peri-centers before the stars become unbound and into their own orbit around the host.

For movies of the simulations, scan the QR code at the top